

ALGOPROG PRE-TEST W UTS BOOTCAMP

Formatted Input / Output:

1. Problem 1: Write a C program that prompts the user to enter their name and age. Display their name and age in the format: "Hello, [Name]! You are [Age] years old."

Numeric Constraint: Age > 0

Input:

Enter your name: Alice

Enter your age: 30

Output:

Hello, Alice! You are 30 years old.

2. Problem 2: Create a program that converts a given temperature in Celsius to Fahrenheit. Prompt the user to enter the temperature in Celsius and display the result in Fahrenheit.

Numeric Constraint: Temperature $\in \mathbb{Z}$

Input:

Enter the temperature in Celsius: 25

Output:

25°C is equal to 77°F.

3. Problem 3: Write a program that calculates and displays the area of a circle. Prompt the user to enter the radius of the circle.

Numeric Constraint: Radius > 0

Input:

Enter the radius of the circle: 7

Output:

The area of the circle with a radius of 7 units is 153.94 square units.

Operator, Operand, and Arithmetic:

1. Problem 1: Write a C program that calculates and displays the sum of two integers entered by the user.

Numeric Constraint: Integer1, Integer2 $\in \mathbb{Z}$

Input:

Enter the first integer: 15

Enter the second integer: 20

Output:

The sum of 15 and 20 is 35.

2. Problem 2: Create a program that takes a number as input and checks if it's even or odd. Display an appropriate message based on the input.

Numeric Constraint: Number $\in \mathbb{Z}$

Input:

Enter a number: 13

Output:

13 is an odd number.

3. Problem 3: Develop a program that calculates the factorial of a positive integer. Prompt the user to enter the integer.

Numeric Constraint: Input > 0

Input:

Enter a positive integer: 5

Output:

The factorial of 5 is 120.

Program Control: Selection:

1. Problem 1: Write a program that takes a year as input and checks if it's a leap year or not. Display a message indicating whether it is a leap year.

Numeric Constraint: $\text{Year} > 0$

Input:

Enter a year: 2024

Output:

2024 is a leap year.

2. Problem 2: Create a program that takes a character as input and checks if it's an uppercase letter, a lowercase letter, or a digit.

Numeric Constraint: $\text{Character} \in \text{ASCII}$

Input:

Enter a character: 5

Output:

5 is a digit.

3. Problem 3: Write a program to calculate the income tax for a given income based on the following conditions:

- $\text{Income} \leq 250,000$: No tax
- $250,000 < \text{Income} \leq 500,000$: 10% tax
- $500,000 < \text{Income} \leq 1,000,000$: 20% tax
- $\text{Income} > 1,000,000$: 30% tax

Numeric Constraint: $\text{Income} > 0$

Input:

Enter your income: 750,000

Output:

You owe 150,000 in taxes (20% of your income).

Program Control: Repetition:

1. Problem 1: Create a program that calculates the sum of all even numbers between 1 and 100.

Output:

Sum of even numbers from 1 to 100 is 2550.

2. Problem 2: Write a program to print the Fibonacci sequence up to a specified number of terms. Prompt the user for the number of terms.

Numeric Constraint : $n > 0$.

Input:

Enter the number of terms: 5

Output:

Fibonacci Sequence: 0 1 1 2 3

Input:

Enter the number of terms: 10

Output:

Fibonacci Sequence: 0 1 1 2 3 5 8 13 21 34

3. Problem 3: Develop a program that calculates the average of a list of numbers entered by the user. Terminate input when the user enters a negative number.

Numeric Constraint: Input $\in \mathbb{Z}$

Input:

Enter a number (enter a negative number to stop): 5

Enter a number (enter a negative number to stop): 8

Enter a number (enter a negative number to stop): -1

Output:

Average of the entered numbers: 6.5

Pointers and Arrays:

1. Problem 1: Write a program that initializes an array of integers and finds the sum of its elements.

Numeric Constraint: $\text{Array}[i] \in \mathbb{Z}, i \in [0, n-1]$

Input:

Enter the number of elements in the array: 5

Enter 5 integers:

10

20

30

40

50

Output:

Sum of array elements: 150

2. Problem 2: Create a program that reads a string from the user and counts the number of vowels in it.

Input:

Enter a string: Hello, World!

Output:

Number of vowels in the string: 3

3. Problem 3: Develop a program that checks if a given string is a palindrome (reads the same backward).

Input:

Enter a string: radar

Output:

The string "radar" is a palindrome.